```
!pip install transformers torch pandas
!pip install transformers datasets accelerate
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12
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Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-p
```

```
import pandas as pd
from transformers import pipeline

# ------------------------------
# Load your dataset
# ------------------------------
df = pd.read_csv("/content/final_2600_dataset.csv")
```

```
# ------------------------------
# Load sentiment model
# ------------------------------
classifier = pipeline("sentiment-analysis")

# ------------------------------
# Generate sentiments
# ------------------------------
# (handles all rows)
results = classifier(df["dialogue"].tolist())

# ------------------------------
# Add sentiment column
# ------------------------------
df["sentiment"] = [r["label"].upper() for r in results]

# ------------------------------
# Optional: Normalize labels to 3 classes
# ------------------------------
def normalize_sentiment(label):
    if "POS" in label:
        return "POSITIVE"
    elif "NEG" in label:
        return "NEGATIVE"
    else:
        return "NEUTRAL"

df["sentiment"] = df["sentiment"].apply(normalize_sentiment)

# ------------------------------
# Save updated dataset
# ------------------------------
df.to_csv("dataset_with_sentiment.csv", index=False)

# ------------------------------
# Show result
# ------------------------------
print("Done ✅")
print("Shape:", df.shape)
df.head()
```

```
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetu
Using a pipeline without specifying a model name and revision in production i
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: Us
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings ta
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access p
```

```
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
```

```
config.json:   0%|          | 0.00/629 [00:00<?, ?B/s]
```

```
model.safetensors:   0%|          | 0.00/268M [00:00<?, ?B/s]
```

```
Loading weights:   0%|          | 0/104 [00:00<?, ?it/s]
```

```
tokenizer_config.json:   0%|          | 0.00/48.0 [00:00<?, ?B/s]
```

```
vocab.txt: 0.00B [00:00, ?B/s]
```

```
Done ✅
Shape: (2600, 3)
```

	character	dialogue	sentiment
0	ANATOLY	We are here.	POSITIVE
1	ANATOLY	5 MINUTES LATER: THE TWO SUBS skim over the se...	NEGATIVE
2	ANATOLY	sidescan sonar and the THRUM of big thrusters.	NEGATIVE
3	ANATOLY	6 The featureless gray clay of the bottom unro...	NEGATIVE
4	ANATOLY	Bodine is watching the sidescan sonar display,...	NEGATIVE

```
<div class="colab-df-buttons">
```

```
<button class="colab-df-convert" onclick="convertToInteractive('df-983cb5c3-f
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>

import pandas as pd
from sklearn.preprocessing import LabelEncoder

# ------------------------------
# Load dataset with sentiment
# ------------------------------
df = pd.read_csv("dataset_with_sentiment.csv")

# ------------------------------
# Encode labels
# ------------------------------
le = LabelEncoder()
df["label"] = le.fit_transform(df["sentiment"])

# ------------------------------
# Save updated dataset
# ------------------------------
df.to_csv("dataset_encoded.csv", index=False)

# ------------------------------
# Check mapping
# ------------------------------
print(dict(zip(le.classes_, le.transform(le.classes_))))
df.head()

{'NEGATIVE': np.int64(0), 'POSITIVE': np.int64(1)}
```

	character	dialogue	sentiment	label
0	ANATOLY	We are here.	POSITIVE	1
1	ANATOLY	5 MINUTES LATER: THE TWO SUBS skim over the se…	NEGATIVE	0
2	ANATOLY	sidescan sonar and the THRUM of big thrusters.	NEGATIVE	0
3	ANATOLY	6 The featureless gray clay of the bottom unro…	NEGATIVE	0
4	ANATOLY	Bodine is watching the sidescan sonar display,…	NEGATIVE	0

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-5448aee1-a
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>

from sklearn.preprocessing import LabelEncoder

# check columns first
print(df.columns)
```

```
# create label column
le = LabelEncoder()
df["label"] = le.fit_transform(df["sentiment"])

# verify
print(df.head())
```

```
Index(['character', 'dialogue', 'sentiment', 'label'], dtype='object')
  character                                          dialogue sentiment  \
0   ANATOLY                                       We are here.  POSITIVE
1   ANATOLY  5 MINUTES LATER: THE TWO SUBS skim over the se...  NEGATIVE
2   ANATOLY     sidescan sonar and the THRUM of big thrusters.  NEGATIVE
3   ANATOLY  6 The featureless gray clay of the bottom unro...  NEGATIVE
4   ANATOLY  Bodine is watching the sidescan sonar display,...  NEGATIVE

   label
0      1
1      0
2      0
3      0
4      0
```

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

# ------------------------------
# STEP 1: Load dataset
# ------------------------------
df = pd.read_csv("/content/dataset_with_sentiment.csv")

# ------------------------------
# STEP 2: Check columns
# ------------------------------
print("Columns:", df.columns)

# ------------------------------
# STEP 3: Fix column names (strip spaces)
# ------------------------------
df.columns = df.columns.str.strip()

# ------------------------------
# STEP 4: Check again
# ------------------------------
print("Fixed Columns:", df.columns)

# ------------------------------
# STEP 5: Ensure 'sentiment' exists
# ------------------------------
```

```
if "sentiment" not in df.columns:
    raise Exception("❌ 'sentiment' column not found. Check file.")

# ------------------------------
# STEP 6: Create label column
# ------------------------------
le = LabelEncoder()
df["label"] = le.fit_transform(df["sentiment"])

print("Label column created ✅")

# ------------------------------
# STEP 7: Split dataset
# ------------------------------
X = df["dialogue"]
y = df["label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Train size:", len(X_train))
print("Test size:", len(X_test))
```

```
Columns: Index(['character', 'dialogue', 'sentiment'], dtype='object')
Fixed Columns: Index(['character', 'dialogue', 'sentiment'], dtype='object')
Label column created ✅
Train size: 2080
Test size: 520
```

```
from datasets import Dataset

# ------------------------------
# Convert pandas Series to Hugging Face Dataset format
# ------------------------------
train_dataset = Dataset.from_dict({
    "text": list(X_train),
    "label": list(y_train)
})

test_dataset = Dataset.from_dict({
    "text": list(X_test),
    "label": list(y_test)
})

print("Train and test datasets created ✅")
```

```
Train and test datasets created ✅
```

```
import numpy as np
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_

def evaluate_model(y_true, y_pred, name):
    accuracy = accuracy_score(y_true, y_pred)
    precision = precision_score(y_true, y_pred, average="weighted", zero_divis
    recall = recall_score(y_true, y_pred, average="weighted", zero_division=0
    f1 = f1_score(y_true, y_pred, average="weighted", zero_division=0)
    mcc = matthews_corrcoef(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    cm = confusion_matrix(y_true, y_pred)

    print(f"\n{name} RESULTS")
    print("Accuracy:", accuracy)
    print("Precision:", precision)
    print("Recall:", recall)
    print("F1 Score:", f1)
    print("MCC:", mcc)
    print("RMSE:", rmse)
    print("Confusion Matrix:\n", cm)
```

```
!pip install transformers datasets accelerate
```

```
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Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-pac
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Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dis
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.1
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.1
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python
```

```
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/pyth
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-pa
```

BART MODEL

```
from transformers import AutoModelForSequenceClassification, Trainer, Training

# ------------------------------
# Load BART model
# ------------------------------
model = AutoModelForSequenceClassification.from_pretrained(
    "facebook/bart-base",
    num_labels=3
)

# ------------------------------
# Load BART Tokenizer
# ------------------------------
tokenizer = AutoTokenizer.from_pretrained("facebook/bart-base")

# ------------------------------
# Tokenize datasets
# ------------------------------
def tokenize_function(examples):
    return tokenizer(examples["text"], truncation=True, padding="max_length")

tokenized_train_dataset = train_dataset.map(tokenize_function, batched=True)
tokenized_test_dataset = test_dataset.map(tokenize_function, batched=True)

# ------------------------------
# Data collator (FIX)
# ------------------------------
data_collator = DataCollatorWithPadding(tokenizer=tokenizer)

# ------------------------------
# Training arguments
# ------------------------------
training_args = TrainingArguments(
    output_dir="./bart_results_3",
    num_train_epochs=3,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./bart_logs_3",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_train_dataset, # Use tokenized dataset
    eval_dataset=tokenized_test_dataset,   # Use tokenized dataset
    data_collator=data_collator
)

# ----------------------------
# Train
# ----------------------------
trainer.train()

# ----------------------------
# Predictions
# ----------------------------
y_pred_bart_3 = trainer.predict(tokenized_test_dataset).predictions[0].argmax

print("BART (3 epochs) DONE ✅")
```

```
config.json: 0.00B [00:00, ?B/s]



model.safetensors:   0%|          | 0.00/558M [00:00<?, ?B/s]



Loading weights:   0%|          | 0/259 [00:00<?, ?it/s]


BartForSequenceClassification LOAD REPORT from: facebook/bart-base
Key                                  | Status  |
-------------------------------------+---------+-
classification_head.dense.weight     | MISSING |
classification_head.out_proj.bias    | MISSING |
classification_head.out_proj.weight  | MISSING |
classification_head.dense.bias       | MISSING |

Notes:
- MISSING   :those params were newly initialized because missing from the che



vocab.json: 0.00B [00:00, ?B/s]
```

```
merges.txt: 0.00B [00:00, ?B/s]


tokenizer.json: 0.00B [00:00, ?B/s]


Map:   0%|          | 0/2080 [00:00<?, ? examples/s]


Map:   0%|          | 0/520 [00:00<?, ? examples/s]

`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='780' max='780' style='width:300px; height:20px; vertical-a
  [780/780 01:31, Epoch 3/3]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.543354

BART (3 epochs) DONE ✅

for 5 epochs

```
from transformers import DataCollatorWithPadding

data_collator = DataCollatorWithPadding(tokenizer=tokenizer)

from transformers import AutoModelForSequenceClassification, Trainer, Training

# ------------------------------
# Reload BART model (fresh training)
# ------------------------------
model = AutoModelForSequenceClassification.from_pretrained(
    "facebook/bart-base",
    num_labels=3
)

# ------------------------------
# Training arguments (5 epochs)
```

```
# ------------------------------
training_args = TrainingArguments(
    output_dir="./bart_results_5",
    num_train_epochs=5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./bart_logs_5",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_train_dataset, # Use tokenized dataset
    eval_dataset=tokenized_test_dataset,   # Use tokenized dataset
    data_collator=data_collator
)

# ------------------------------
# Train
# ------------------------------
trainer.train()

print("BART (5 epochs) DONE ✅")
```

```
Loading weights:   0%|          | 0/259 [00:00<?, ?it/s]


BartForSequenceClassification LOAD REPORT from: facebook/bart-base
Key                                  | Status  |
-------------------------------------+---------+-
classification_head.dense.weight     | MISSING |
classification_head.out_proj.bias    | MISSING |
classification_head.out_proj.weight  | MISSING |
classification_head.dense.bias       | MISSING |

Notes:
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
  <progress value='1300' max='1300' style='width:300px; height:20px; vertical
  [1300/1300 02:20, Epoch 5/5]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.530650 1000 0.183789

BART (5 epochs) DONE ✅

```
y_pred_bart_5 = trainer.predict(tokenized_test_dataset).predictions[0].argmax
```

BART with 7 Epochs

```
from transformers import AutoModelForSequenceClassification, Trainer, Training

# ------------------------------
# Reload BART model (fresh training)
# ------------------------------
model = AutoModelForSequenceClassification.from_pretrained(
    "facebook/bart-base",
    num_labels=3
)

# ------------------------------
# Training arguments (7 epochs)
# ------------------------------
training_args = TrainingArguments(
    output_dir="./bart_results_7",
    num_train_epochs=7,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./bart_logs_7",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_train_dataset,
    eval_dataset=tokenized_test_dataset,
    data_collator=data_collator
)

# ------------------------------
# Train
```

```
# ------------------------------
trainer.train()

print("BART (7 epochs) DONE ✅")
```

```
Loading weights:   0%|          | 0/259 [00:00<?, ?it/s]


BartForSequenceClassification LOAD REPORT from: facebook/bart-base
Key                                   | Status  |
--------------------------------------+---------+-
classification_head.dense.weight      | MISSING |
classification_head.out_proj.bias     | MISSING |
classification_head.out_proj.weight   | MISSING |
classification_head.dense.bias        | MISSING |

Notes:
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='1820' max='1820' style='width:300px; height:20px; vertical
  [1820/1820 02:51, Epoch 7/7]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.562924 1000 0.265496 1500 0.116124

```
BART (7 epochs) DONE ✅

y_pred_bart_7 = trainer.predict(tokenized_test_dataset).predictions[0].argmax
```

BART Results

```
evaluate_model(y_test, y_pred_bart_3, "BART - 3 Epochs")
evaluate_model(y_test, y_pred_bart_5, "BART - 5 Epochs")
evaluate_model(y_test, y_pred_bart_7, "BART - 7 Epochs")
```

```
BART - 3 Epochs RESULTS
Accuracy: 0.7711538461538462
Precision: 0.7728735207100591
Recall: 0.7711538461538462
F1 Score: 0.7692001098109273
MCC: 0.5387637541291109
```

```
RMSE: 0.47837867202264966
Confusion Matrix:
 [[164  75]
 [ 44 237]]

BART - 5 Epochs RESULTS
Accuracy: 0.7980769230769231
Precision: 0.7982894161585437
Recall: 0.7980769230769231
F1 Score: 0.7973181578854086
MCC: 0.5926282567542644
RMSE: 0.4493585171364586
Confusion Matrix:
 [[178  61]
 [ 44 237]]

BART - 7 Epochs RESULTS
Accuracy: 0.7769230769230769
Precision: 0.7767527162103433
Recall: 0.7769230769230769
F1 Score: 0.7762686965811966
MCC: 0.5497571051179371
RMSE: 0.4723101979387308
Confusion Matrix:
 [[174  65]
 [ 51 230]]
```

DeBERTa Model Training

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification, T

# ------------------------------
# Load DeBERTa Tokenizer
# ------------------------------
model_name_deberta = "microsoft/deberta-base"
tokenizer_deberta = AutoTokenizer.from_pretrained(model_name_deberta)

def tokenize_deberta(example):
    return tokenizer_deberta(example["text"], truncation=True, padding="max_le

train_dataset_deberta = train_dataset.map(tokenize_deberta, batched=True)
test_dataset_deberta = test_dataset.map(tokenize_deberta, batched=True)

# ------------------------------
# Data collator
# ------------------------------
data_collator_deberta = DataCollatorWithPadding(tokenizer=tokenizer_deberta)
```

```
print("DeBERTa tokenizer and data collator loaded ✅")
```

```
config.json:   0%|          | 0.00/474 [00:00<?, ?B/s]


tokenizer_config.json:   0%|          | 0.00/52.0 [00:00<?, ?B/s]


vocab.json: 0.00B [00:00, ?B/s]


merges.txt: 0.00B [00:00, ?B/s]


Map:   0%|          | 0/2080 [00:00<?, ? examples/s]


Map:   0%|          | 0/520 [00:00<?, ? examples/s]

DeBERTa tokenizer and data collator loaded ✅
```

DeBERTa with 3 Epochs

```
# ------------------------------
# Load DeBERTa model
# ------------------------------
model_deberta_3 = AutoModelForSequenceClassification.from_pretrained(
    model_name_deberta,
    num_labels=3
)

# ------------------------------
# Training arguments (3 epochs)
# ------------------------------
training_args_deberta_3 = TrainingArguments(
    output_dir="./deberta_results_3",
    num_train_epochs=3,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./deberta_logs_3",
    save_strategy="no",
    fp16=False # Disable mixed precision training to prevent overflow
```

```
)

# ------------------------------
# Trainer
# ------------------------------
trainer_deberta_3 = Trainer(
    model=model_deberta_3,
    args=training_args_deberta_3,
    train_dataset=train_dataset_deberta,
    eval_dataset=test_dataset_deberta,
    data_collator=data_collator_deberta
)

# ------------------------------
# Train
# ------------------------------
trainer_deberta_3.train()

# ------------------------------
# Predictions
# ------------------------------
y_pred_deberta_3 = trainer_deberta_3.predict(test_dataset_deberta).predictions

print("DeBERTa (3 epochs) DONE ✅")
```

```
Loading weights:   0%|          | 0/196 [00:00<?, ?it/s]


DebertaForSequenceClassification LOAD REPORT from: microsoft/deberta-base
Key                                        | Status     |
-------------------------------------------+------------+-
lm_predictions.lm_head.dense.bias          | UNEXPECTED |
lm_predictions.lm_head.LayerNorm.bias      | UNEXPECTED |
lm_predictions.lm_head.bias                | UNEXPECTED |
lm_predictions.lm_head.dense.weight        | UNEXPECTED |
lm_predictions.lm_head.LayerNorm.weight    | UNEXPECTED |
pooler.dense.weight                        | MISSING    |
classifier.bias                            | MISSING    |
classifier.weight                          | MISSING    |
pooler.dense.bias                          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='780' max='780' style='width:300px; height:20px; vertical-a
  [780/780 01:32, Epoch 3/3]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.566298

DeBERTa (3 epochs) DONE ✅

DeBERTa with 5 Epochs

```
# ------------------------------
# Reload DeBERTa model (fresh training)
# ------------------------------
model_deberta_5 = AutoModelForSequenceClassification.from_pretrained(
    model_name_deberta,
    num_labels=3
)

# ------------------------------
# Training arguments (5 epochs)
# ------------------------------
training_args_deberta_5 = TrainingArguments(
    output_dir="./deberta_results_5",
    num_train_epochs=5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./deberta_logs_5",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_deberta_5 = Trainer(
    model=model_deberta_5,
    args=training_args_deberta_5,
    train_dataset=train_dataset_deberta,
    eval_dataset=test_dataset_deberta,
    data_collator=data_collator_deberta
)

# ------------------------------
# Train
# ------------------------------
```

```
trainer_deberta_5.train()

# ------------------------------
# Predictions
# ------------------------------
y_pred_deberta_5 = trainer_deberta_5.predict(test_dataset_deberta).predictions

print("DeBERTa (5 epochs) DONE ✅")
```

```
Loading weights:   0%|          | 0/196 [00:00<?, ?it/s]


DebertaForSequenceClassification LOAD REPORT from: microsoft/deberta-base
Key                                       | Status     |
------------------------------------------+------------+-
lm_predictions.lm_head.dense.bias         | UNEXPECTED |
lm_predictions.lm_head.LayerNorm.bias     | UNEXPECTED |
lm_predictions.lm_head.bias               | UNEXPECTED |
lm_predictions.lm_head.dense.weight       | UNEXPECTED |
lm_predictions.lm_head.LayerNorm.weight   | UNEXPECTED |
pooler.dense.weight                       | MISSING    |
classifier.bias                           | MISSING    |
classifier.weight                         | MISSING    |
pooler.dense.bias                         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='1300' max='1300' style='width:300px; height:20px; vertical
  [1300/1300 02:33, Epoch 5/5]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.605407 1000 0.283095

```
DeBERTa (5 epochs) DONE ✅
```

DeBERTa with 7 Epochs

```
# ------------------------------
# Reload DeBERTa model (fresh training)
```

```
# ------------------------------
model_deberta_7 = AutoModelForSequenceClassification.from_pretrained(
    model_name_deberta,
    num_labels=3
)

# ------------------------------
# Training arguments (7 epochs)
# ------------------------------
training_args_deberta_7 = TrainingArguments(
    output_dir="./deberta_results_7",
    num_train_epochs=7,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./deberta_logs_7",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_deberta_7 = Trainer(
    model=model_deberta_7,
    args=training_args_deberta_7,
    train_dataset=train_dataset_deberta,
    eval_dataset=test_dataset_deberta,
    data_collator=data_collator_deberta
)

# ------------------------------
# Train
# ------------------------------
trainer_deberta_7.train()

# ------------------------------
# Predictions
# ------------------------------
y_pred_deberta_7 = trainer_deberta_7.predict(test_dataset_deberta).predictions

print("DeBERTa (7 epochs) DONE ✅")
```

```
Loading weights:   0%|          | 0/196 [00:00<?, ?it/s]


DebertaForSequenceClassification LOAD REPORT from: microsoft/deberta-base
Key                                       | Status     |
------------------------------------------+------------+-
lm_predictions.lm_head.dense.bias         | UNEXPECTED |
```

```
lm_predictions.lm_head.LayerNorm.bias   | UNEXPECTED |
lm_predictions.lm_head.bias             | UNEXPECTED |
lm_predictions.lm_head.dense.weight     | UNEXPECTED |
lm_predictions.lm_head.LayerNorm.weight | UNEXPECTED |
pooler.dense.weight                     | MISSING    |
classifier.bias                         | MISSING    |
classifier.weight                       | MISSING    |
pooler.dense.bias                       | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='1820' max='1820' style='width:300px; height:20px; vertical
  [1820/1820 03:32, Epoch 7/7]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.580475 1000 0.253544 1500 0.062540

DeBERTa (7 epochs) DONE ✅

DeBERTa Results

```
evaluate_model(y_test, y_pred_deberta_3, "DeBERTa - 3 Epochs")
evaluate_model(y_test, y_pred_deberta_5, "DeBERTa - 5 Epochs")
evaluate_model(y_test, y_pred_deberta_7, "DeBERTa - 7 Epochs")
```

```
DeBERTa - 3 Epochs RESULTS
Accuracy: 0.7923076923076923
Precision: 0.7943420431320245
Recall: 0.7923076923076923
F1 Score: 0.7906161828036827
MCC: 0.581941864211242
RMSE: 0.45573271518765
Confusion Matrix:
 [[170  69]
 [ 39 242]]

DeBERTa - 5 Epochs RESULTS
Accuracy: 0.8096153846153846
Precision: 0.8122904339250493
```

```
Recall: 0.8096153846153846
F1 Score: 0.8079900073216958
MCC: 0.6175303848497411
RMSE: 0.4363308554120547
Confusion Matrix:
 [[174  65]
 [ 34 247]]

DeBERTa - 7 Epochs RESULTS
Accuracy: 0.7846153846153846
Precision: 0.7854599841637585
Recall: 0.7846153846153846
F1 Score: 0.7833330056451044
MCC: 0.5656075007013708
RMSE: 0.4640954808922571
Confusion Matrix:
 [[171  68]
 [ 44 237]]
```

DistilBERT Model Training

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification, T

# ------------------------------
# Load DistilBERT Tokenizer
# ------------------------------
model_name_distilbert = "distilbert-base-uncased"
tokenizer_distilbert = AutoTokenizer.from_pretrained(model_name_distilbert)

def tokenize_distilbert(example):
    return tokenizer_distilbert(example["text"], truncation=True, padding="ma

train_dataset_distilbert = train_dataset.map(tokenize_distilbert, batched=Tru
test_dataset_distilbert = test_dataset.map(tokenize_distilbert, batched=True)

# ------------------------------
# Data collator
# ------------------------------
data_collator_distilbert = DataCollatorWithPadding(tokenizer=tokenizer_distil

print("DistilBERT tokenizer and data collator loaded ✅")
```

```
config.json:   0%|          | 0.00/483 [00:00<?, ?B/s]



tokenizer_config.json:   0%|          | 0.00/48.0 [00:00<?, ?B/s]
```

```
vocab.txt: 0.00B [00:00, ?B/s]


tokenizer.json: 0.00B [00:00, ?B/s]


Map:   0%|          | 0/2080 [00:00<?, ? examples/s]


Map:   0%|          | 0/520 [00:00<?, ? examples/s]

DistilBERT tokenizer and data collator loaded ✅
```

DistilBERT with 3 Epochs

```
# ------------------------------
# Load DistilBERT model
# ------------------------------
model_distilbert_3 = AutoModelForSequenceClassification.from_pretrained(
    model_name_distilbert,
    num_labels=3
)

# ------------------------------
# Training arguments (3 epochs)
# ------------------------------
training_args_distilbert_3 = TrainingArguments(
    output_dir="./distilbert_results_3",
    num_train_epochs=3,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./distilbert_logs_3",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_distilbert_3 = Trainer(
    model=model_distilbert_3,
    args=training_args_distilbert_3,
    train_dataset=train_dataset_distilbert,
```

```
    eval_dataset=test_dataset_distilbert,
    data_collator=data_collator_distilbert
)

# ----------------------------
# Train
# ----------------------------
trainer_distilbert_3.train()

# ----------------------------
# Predictions
# ----------------------------
y_pred_distilbert_3 = trainer_distilbert_3.predict(test_dataset_distilbert).p

print("DistilBERT (3 epochs) DONE ✅")
```

```
model.safetensors:   0%|          | 0.00/268M [00:00<?, ?B/s]



Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]


DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
classifier.bias          | MISSING    |
classifier.weight        | MISSING    |
pre_classifier.weight    | MISSING    |
pre_classifier.bias      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='780' max='780' style='width:300px; height:20px; vertical-a
  [780/780 05:18, Epoch 3/3]
```

```
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.409355

```
DistilBERT (3 epochs) DONE ✅
```

DistilBERT with 5 Epochs

```
# ------------------------------
# Reload DistilBERT model (fresh training)
# ------------------------------
model_distilbert_5 = AutoModelForSequenceClassification.from_pretrained(
    model_name_distilbert,
    num_labels=3
)

# ------------------------------
# Training arguments (5 epochs)
# ------------------------------
training_args_distilbert_5 = TrainingArguments(
    output_dir="./distilbert_results_5",
    num_train_epochs=5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./distilbert_logs_5",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_distilbert_5 = Trainer(
    model=model_distilbert_5,
    args=training_args_distilbert_5,
    train_dataset=train_dataset_distilbert,
    eval_dataset=test_dataset_distilbert,
    data_collator=data_collator_distilbert
)

# ------------------------------
# Train
# ------------------------------
trainer_distilbert_5.train()

# ------------------------------
# Predictions
# ------------------------------
```

```
y_pred_distilbert_5 = trainer_distilbert_5.predict(test_dataset_distilbert).p

print("DistilBERT (5 epochs) DONE ✅")
```

```
Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]


DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
classifier.bias          | MISSING    |
classifier.weight        | MISSING    |
pre_classifier.weight    | MISSING    |
pre_classifier.bias      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='1300' max='1300' style='width:300px; height:20px; vertical
  [1300/1300 08:47, Epoch 5/5]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.421362 1000 0.056449

```
DistilBERT (5 epochs) DONE ✅
```

DistilBERT with 7 Epochs

```
# ------------------------------
# Reload DistilBERT model (fresh training)
# ------------------------------
model_distilbert_7 = AutoModelForSequenceClassification.from_pretrained(
    model_name_distilbert,
    num_labels=3
)
```

```
# ------------------------------
# Training arguments (7 epochs)
# ------------------------------
training_args_distilbert_7 = TrainingArguments(
    output_dir="./distilbert_results_7",
    num_train_epochs=7,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./distilbert_logs_7",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_distilbert_7 = Trainer(
    model=model_distilbert_7,
    args=training_args_distilbert_7,
    train_dataset=train_dataset_distilbert,
    eval_dataset=test_dataset_distilbert,
    data_collator=data_collator_distilbert
)

# ------------------------------
# Train
# ------------------------------
trainer_distilbert_7.train()

# ------------------------------
# Predictions
# ------------------------------
y_pred_distilbert_7 = trainer_distilbert_7.predict(test_dataset_distilbert).p

print("DistilBERT (7 epochs) DONE ✅")
```

```
Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]


DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
classifier.bias          | MISSING    |
```

```
classifier.weight       | MISSING    |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture
- MISSING   :those params were newly initialized because missing from the che
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBO
```

```
<div>

  <progress value='1820' max='1820' style='width:300px; height:20px; vertical
  [1820/1820 12:18, Epoch 7/7]
</div>
<table border="1" class="dataframe">
```

Step Training Loss 500 0.438658 1000 0.055696 1500 0.011871

```
DistilBERT (7 epochs) DONE ✅
```

DistilBERT Results

```
evaluate_model(y_test, y_pred_distilbert_3, "DistilBERT - 3 Epochs")
evaluate_model(y_test, y_pred_distilbert_5, "DistilBERT - 5 Epochs")
evaluate_model(y_test, y_pred_distilbert_7, "DistilBERT - 7 Epochs")
```

```
DistilBERT - 3 Epochs RESULTS
Accuracy: 0.8403846153846154
Precision: 0.8407813892188891
Recall: 0.8403846153846154
F1 Score: 0.8398737763690544
MCC: 0.6783142721582107
RMSE: 0.3995189414976274
Confusion Matrix:
 [[190  49]
 [ 34 247]]

DistilBERT - 5 Epochs RESULTS
Accuracy: 0.8211538461538461
Precision: 0.8236296230651069
Recall: 0.8211538461538461
F1 Score: 0.8197661252520896
MCC: 0.6407901971919611
RMSE: 0.42290206176626033
Confusion Matrix:
```

```
 [[178  61]
 [ 32 249]]

DistilBERT - 7 Epochs RESULTS
Accuracy: 0.8269230769230769
Precision: 0.8268009383926338
Recall: 0.8269230769230769
F1 Score: 0.826664656731846
MCC: 0.6510132237896609
RMSE: 0.41602514716892186
Confusion Matrix:
 [[190  49]
 [ 41 240]]
```

SVM Models with Various Kernels and Voting Techniques

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.svm import SVC
from sklearn.ensemble import VotingClassifier
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

print("Libraries loaded ✅")
```

```
Libraries loaded ✅
```

```
# ------------------------------
# TF-IDF Vectorization
# ------------------------------
vectorizer = TfidfVectorizer(max_features=5000) # Limiting features to avoid o
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

print("TF-IDF Vectorization complete ✅")
print("Shape of X_train_tfidf:", X_train_tfidf.shape)
print("Shape of X_test_tfidf:", X_test_tfidf.shape)
```

```
TF-IDF Vectorization complete ✅
Shape of X_train_tfidf: (2080, 3868)
Shape of X_test_tfidf: (520, 3868)
```

Individual SVM Models

```
# ------------------------------
# SVM with Linear Kernel
# ------------------------------
print("Training Linear SVM...")
svm_linear = Pipeline([
```

```
    ('scaler', StandardScaler(with_mean=False)), # with_mean=False for sparse
    ('svm', SVC(kernel='linear', probability=True, random_state=42))
])
svm_linear.fit(X_train_tfidf, y_train)
y_pred_svm_linear = svm_linear.predict(X_test_tfidf)
print("Linear SVM trained ✅")

# ------------------------------
# SVM with RBF Kernel (Gaussian)
# ------------------------------
print("Training RBF SVM...")
svm_rbf = Pipeline([
    ('scaler', StandardScaler(with_mean=False)),
    ('svm', SVC(kernel='rbf', probability=True, random_state=42))
])
svm_rbf.fit(X_train_tfidf, y_train)
y_pred_svm_rbf = svm_rbf.predict(X_test_tfidf)
print("RBF SVM trained ✅")

# ------------------------------
# SVM with Polynomial Kernel
# ------------------------------
print("Training Polynomial SVM...")
svm_poly = Pipeline([
    ('scaler', StandardScaler(with_mean=False)),
    ('svm', SVC(kernel='poly', degree=3, probability=True, random_state=42))
])
svm_poly.fit(X_train_tfidf, y_train)
y_pred_svm_poly = svm_poly.predict(X_test_tfidf)
print("Polynomial SVM trained ✅")
```

```
Training Linear SVM...
Linear SVM trained ✅
Training RBF SVM...
RBF SVM trained ✅
Training Polynomial SVM...
Polynomial SVM trained ✅
```

Voting Classifier with SVMs

```
# ------------------------------
# Voting Classifier
# ------------------------------
print("Training Voting Classifier...")
voting_clf = VotingClassifier(
    estimators=[
        ('linear', svm_linear),
        ('rbf', svm_rbf),
```

```
        ('poly', svm_poly)
    ],
    voting='soft', # Use 'soft' voting if probability=True in SVC
    weights=[1, 1, 1] # Equal weights for simplicity, can be tuned
)
voting_clf.fit(X_train_tfidf, y_train)
y_pred_svm_voting = voting_clf.predict(X_test_tfidf)
print("Voting Classifier trained ✅")
```

```
Training Voting Classifier...
Voting Classifier trained ✅
```

SVM Results

```
evaluate_model(y_test, y_pred_svm_linear, "SVM - Linear Kernel")
evaluate_model(y_test, y_pred_svm_rbf, "SVM - RBF Kernel")
evaluate_model(y_test, y_pred_svm_poly, "SVM - Polynomial Kernel")
evaluate_model(y_test, y_pred_svm_voting, "SVM - Voting Classifier")
```

```
SVM - Linear Kernel RESULTS
Accuracy: 0.6442307692307693
Precision: 0.6441216011772958
Recall: 0.6442307692307693
F1 Score: 0.64417380397306
MCC: 0.2835673204423471
RMSE: 0.596463939202724
Confusion Matrix:
 [[146  93]
 [ 92 189]]

SVM - RBF Kernel RESULTS
Accuracy: 0.5923076923076923
Precision: 0.5943856143856143
Recall: 0.5923076923076923
F1 Score: 0.5674550321609145
MCC: 0.16680431044267674
RMSE: 0.6385078759829887
Confusion Matrix:
 [[ 81 158]
 [ 54 227]]

SVM - Polynomial Kernel RESULTS
Accuracy: 0.5384615384615384
Precision: 0.5041723562433622
Recall: 0.5384615384615384
F1 Score: 0.3976869154534129
MCC: 0.0006178962877914625
RMSE: 0.6793662204867574
```

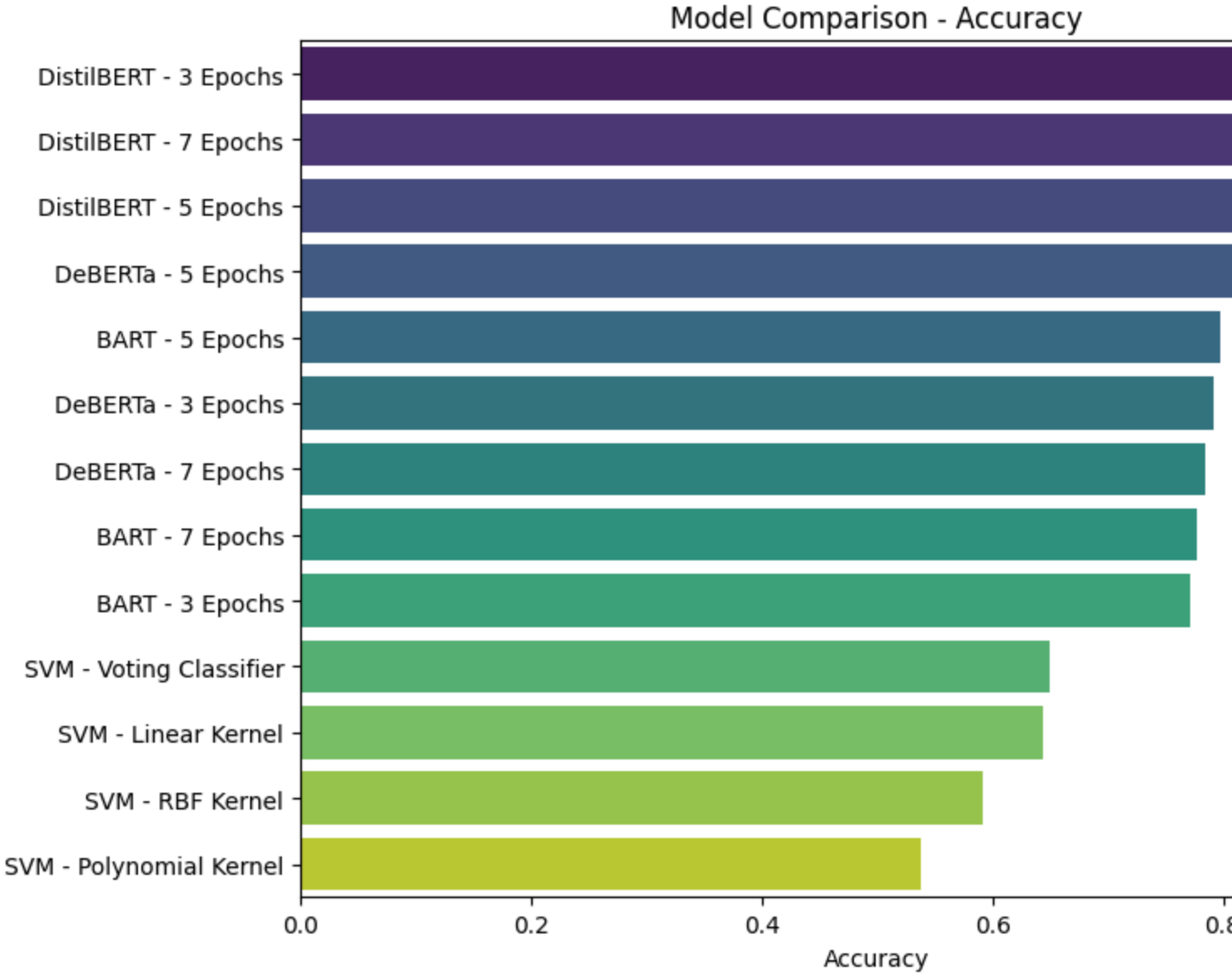
```
Confusion Matrix:
 [[  6 233]
 [  7 274]]

SVM - Voting Classifier RESULTS
Accuracy: 0.65
Precision: 0.6489091078615717
Recall: 0.65
F1 Score: 0.6462817233916979
MCC: 0.29027984987504657
RMSE: 0.5916079783099616
Confusion Matrix:
 [[130 109]
 [ 73 208]]
```

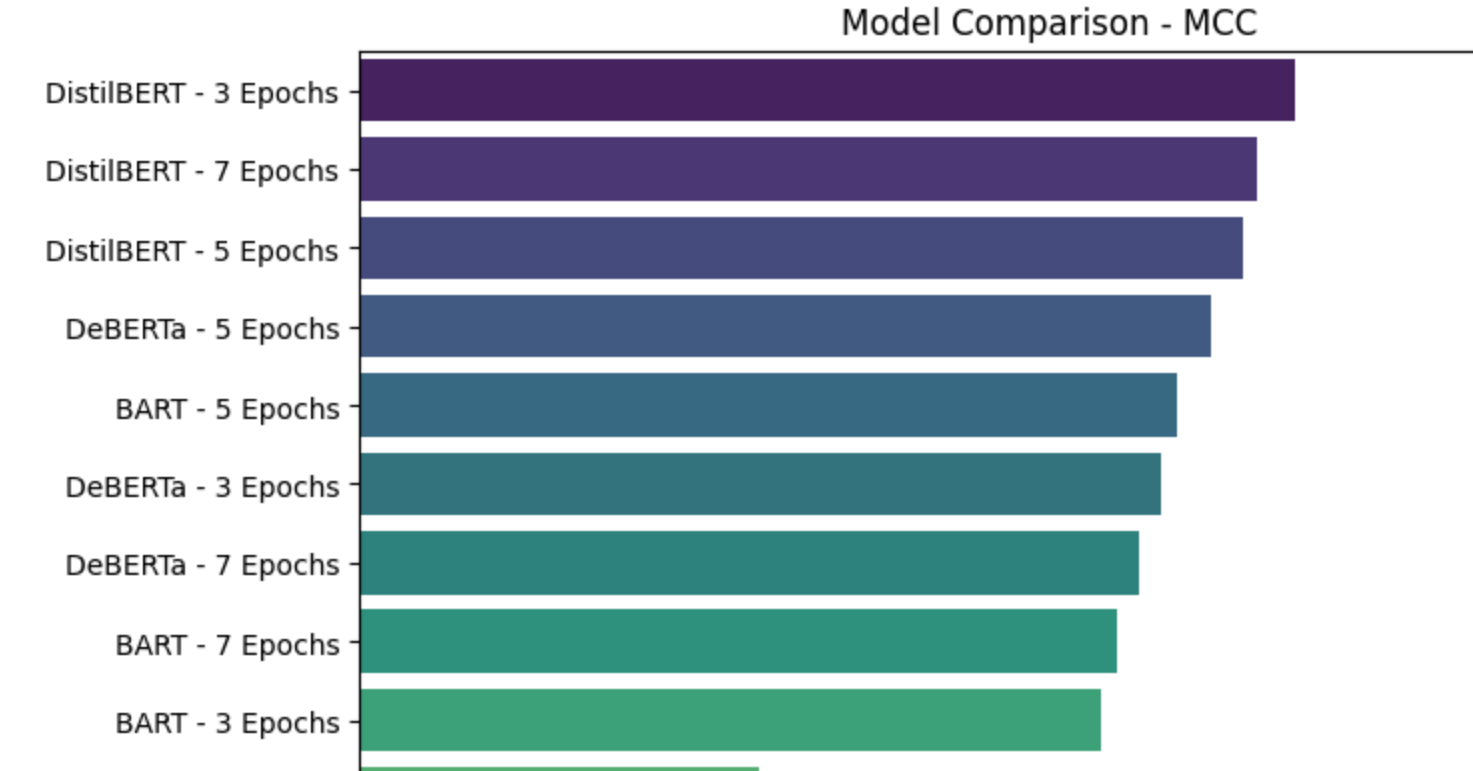
Model Comparison

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Data for BART models
bart_data = {
    "Model": ["BART - 3 Epochs", "BART - 5 Epochs", "BART - 7 Epochs"],
    "Accuracy": [0.7711538461538462, 0.7980769230769231, 0.7769230769230769],
    "Precision": [0.7728735207100591, 0.7982894161585437, 0.7767527162103433]
    "Recall": [0.7711538461538462, 0.7980769230769231, 0.7769230769230769],
    "F1 Score": [0.7692001098109273, 0.7973181578854086, 0.7762686965811966],
    "MCC": [0.5387637541291109, 0.5926282567542644, 0.5497571051179371],
    "RMSE": [0.47837867202264966, 0.4493585171364586, 0.4723101979387308]
}

# Data for DeBERTa models
deberta_data = {
    "Model": ["DeBERTa - 3 Epochs", "DeBERTa - 5 Epochs", "DeBERTa - 7 Epochs
    "Accuracy": [0.7923076923076923, 0.8096153846153846, 0.7846153846153846],
    "Precision": [0.7943420431320245, 0.8122904339250493, 0.7854599841637585]
    "Recall": [0.7923076923076923, 0.8096153846153846, 0.7846153846153846],
    "F1 Score": [0.7906161828036827, 0.8079900073216958, 0.7833330056451044],
    "MCC": [0.581941864211242, 0.6175303848497411, 0.5656075007013708],
    "RMSE": [0.45573271518765, 0.4363308554120547, 0.4640954808922571]
}

# Data for DistilBERT models
distilbert_data = {
    "Model": ["DistilBERT - 3 Epochs", "DistilBERT - 5 Epochs", "DistilBERT -
    "Accuracy": [0.8403846153846154, 0.8211538461538461, 0.8269230769230769],
    "Precision": [0.8407813892188891, 0.8236296230651069, 0.8268009383926338]
```

```
    "Recall": [0.8403846153846154, 0.8211538461538461, 0.8269230769230769],
    "F1 Score": [0.8398737763690544, 0.8197661252520896, 0.826664656731846],
    "MCC": [0.6783142721582107, 0.6407901971919611, 0.6510132237896609],
    "RMSE": [0.3995189414976274, 0.42290206176626033, 0.41602514716892186]
}

# Data for SVM models
svm_data = {
    "Model": ["SVM - Linear Kernel", "SVM - RBF Kernel", "SVM - Polynomial Ke
    "Accuracy": [0.6442307692307693, 0.5923076923076923, 0.5384615384615384, 0
    "Precision": [0.6441216011772958, 0.5943856143856143, 0.5041723562433622,
    "Recall": [0.6442307692307693, 0.5923076923076923, 0.5384615384615384, 0.0
    "F1 Score": [0.64417380397306, 0.5674550321609145, 0.3976869154534129, 0.0
    "MCC": [0.2835673204423471, 0.16680431044267674, 0.0006178962877914625, 0
    "RMSE": [0.596463939202724, 0.6385078759829887, 0.6793662204867574, 0.5910
}

# Create DataFrames
df_bart = pd.DataFrame(bart_data)
df_deberta = pd.DataFrame(deberta_data)
df_distilbert = pd.DataFrame(distilbert_data)
df_svm = pd.DataFrame(svm_data)

# Combine all results into a single DataFrame
all_results = pd.concat([df_bart, df_deberta, df_distilbert, df_svm], ignore_i

# Sort by Accuracy for better visualization
all_results_sorted = all_results.sort_values(by="Accuracy", ascending=False)

# Melt the DataFrame for easier plotting of multiple metrics
all_results_melted = all_results_sorted.melt(id_vars="Model", var_name="Metric

# Filter for key metrics for visualization
metrics_to_plot = ["Accuracy", "F1 Score", "MCC", "RMSE"]
plot_data = all_results_melted[all_results_melted["Metric"].isin(metrics_to_pl

# Create subplots for each metric
fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(18, 12))
axes = axes.flatten()

for i, metric in enumerate(metrics_to_plot):
    sns.barplot(x="Score", y="Model", hue="Model", data=plot_data[plot_data["M
    axes[i].set_title(f"Model Comparison - {metric}")
    axes[i].set_xlabel(metric)
    axes[i].set_ylabel("")
    axes[i].set_xlim(0, 1) # Metrics are typically between 0 and 1
    if metric == "RMSE":
        axes[i].set_xlim(0, 0.7) # RMSE can be higher than 1 but here is betwe
```

```
plt.tight_layout()
plt.show()

print("Model comparison graphs generated ✅")
```


Model Comparison - Accuracy
DistilBERT - 3 Epochs
DistilBERT - 7 Epochs
DistilBERT - 5 Epochs
DeBERTa - 5 Epochs
BART - 5 Epochs
DeBERTa - 3 Epochs
DeBERTa - 7 Epochs
BART - 7 Epochs
BART - 3 Epochs
SVM - Voting Classifier
SVM - Linear Kernel
SVM - RBF Kernel
SVM - Polynomial Kernel
0.0
0.2
0.4
0.6
Accuracy


Model Comparison - MCC
DistilBERT - 3 Epochs
DistilBERT - 7 Epochs
DistilBERT - 5 Epochs
DeBERTa - 5 Epochs
BART - 5 Epochs
DeBERTa - 3 Epochs
DeBERTa - 7 Epochs
BART - 7 Epochs
BART - 3 Epochs

png

```
Model comparison graphs generated ✅
```